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DECEMBER 2025

SENTIMENT ANALYSIS OF PTT STOCK BOARD COMMENTS

DATA SCIENCE PROJECT

PROJECT OVERVIEW & TOPIC

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APPROACH

- **Sentiment analysis:** Determination of collective sentiment through social media data
- **Applications:** Economics, Marketing, Finance, Politics, Sociology

TEAM ROLES

- **趙珍萱:** Problem definition, literature review, presentation design
- **李岳榮:** Literature Review, Project design, interpretation
- **潘振軒:** Literature Review, Code

OUR PROJECT

- **Method:** Binary text classification
- **Data:** PTT Stock board user comments
- **Tools:** TF-IDF, Logistic Regression, Perceptron, Accuracy, Precision, Recall
- **Objective:** Determine general sentiment towards Taiwan's financial market or particular stocks

text	sentiment
危險了！！	positive
新進場做多!	positive
？？？？？	positive
四一飛衝天	positive

PROBLEM

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Sentiment analysis

- Traditional methods (historical data, financial reports, interviews) cannot anticipate early signs or trends
- Structured data do not capture emotion or expectations
- Provides valuable information about collective preferences and social behavior

Social media

- Valuable resource for data
- Up-to-date information
- Real time-insights
- Abundant and publicly available

- Unstructured information
- Language barriers
- Ambiguous
- Quantity of data
- Noise

Financial markets

- Specially volatile compared to other markets
- Perception changes extremely rapidly after events, especially in sectors like technology
- Sentiment is an important determinant of demand, and therefore of prices

PTT Stock board

- **Taiwan's emerging and speculative market**
- **Representative platform in Taiwan for retail investors**
- **High costs in uninformed decisions**
- **High Demand for Grassroots Analysis**

REAL WORLD PROBLEM?

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GameStop
(2021 - Reddit / r/WallStreetBets)

Bud Light
(2023 - Twitter/X)

Tesla (Always - Twitter/X)

In Taiwan, retail investors systematically under-perform the market, loosing approximately 3.8% of their total investment value each year, amounting to around NT$940 billion.

→

"Sentiment analysis of social media data has emerged as a pivotal tool for understanding public opinion and forecasting market trends. " (Choudhary, 2025, p. 1)

Sentiment analysis in combination with traditional tools could grasp robust models to forecast stock market trends. (Picasso et al., 2019)

PROJECT DESIGN

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Data collection and processing

- Fetch PTT webpage
- Preprocess comment content
- Scrape comments from a specified article
- Export text as a plain text file for AI labeling
- Prepare environment for traditional chinese

TF-IDF

- TF-IDF analysis
- TF-IDF results vs. raw count
- TF-IDF Vectorization

Model training

- Perceptron
- Logistic regression

Black-box design

INPUT → PTT Comments (TF-IDF Vectors)

OUTPUT → Prediction

Performance Evaluation

- Accuracy: For total predictions
- Precision: For positive predictions
- F1-Score
- Confusion Matrix

Data Collection

↓

Text

↓

Tokenization

↓

Vectorization

↓

Classifier

↓

Prediction

DATA COLLECTION AND PROCESSING

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1. DATA & SOURCE

- Comments collected from **PTT Stock Board**.
- Total of **371** comments.
- Each comment labeled as positive 推 or negative 噓.

2. DATA COLLECTION

- Web scrapping
- Requests + BeautifulSoup.
- Fetches HTML pages and extracts comments automatically.
- Focused on posts with clear sentiment

3. DATA CLEANING

- Remove colons, newlines, extra spaces, and strange symbols.
- Keep only Chinese, English, punctuation, and spaces.
- Ignore short comments. (< 4 characters).

4. TEXT PROCESSING

- Use jieba to segment Chinese text into words.
- Count word frequency with Counter for analysis.
- Prepare data for visualization.

5. VISUALIZATION PREPARATION

- CJK fonts for Traditional Chinese.
- Ensure Chinese characters display correctly in graphs.

總筆數： 371

	text	sentiment
0	空軍危險了！！	positive
1	明天重新進場做多!	positive
2	？？？？？？？？？？？？？	positive
3	穩了 等著禮拜四一飛衝天	positive
4	傻多們準備公園了	positive
...	...	...
366	這傢伙又翻車了	positive
367	感謝老師	positive
368	謝謝謝國師勇敢加倉有賺到	positive
369	朝聖 謝大師	positive
370	謝天謝地謝金河	positive

371 rows × 2 columns

TF-IDF

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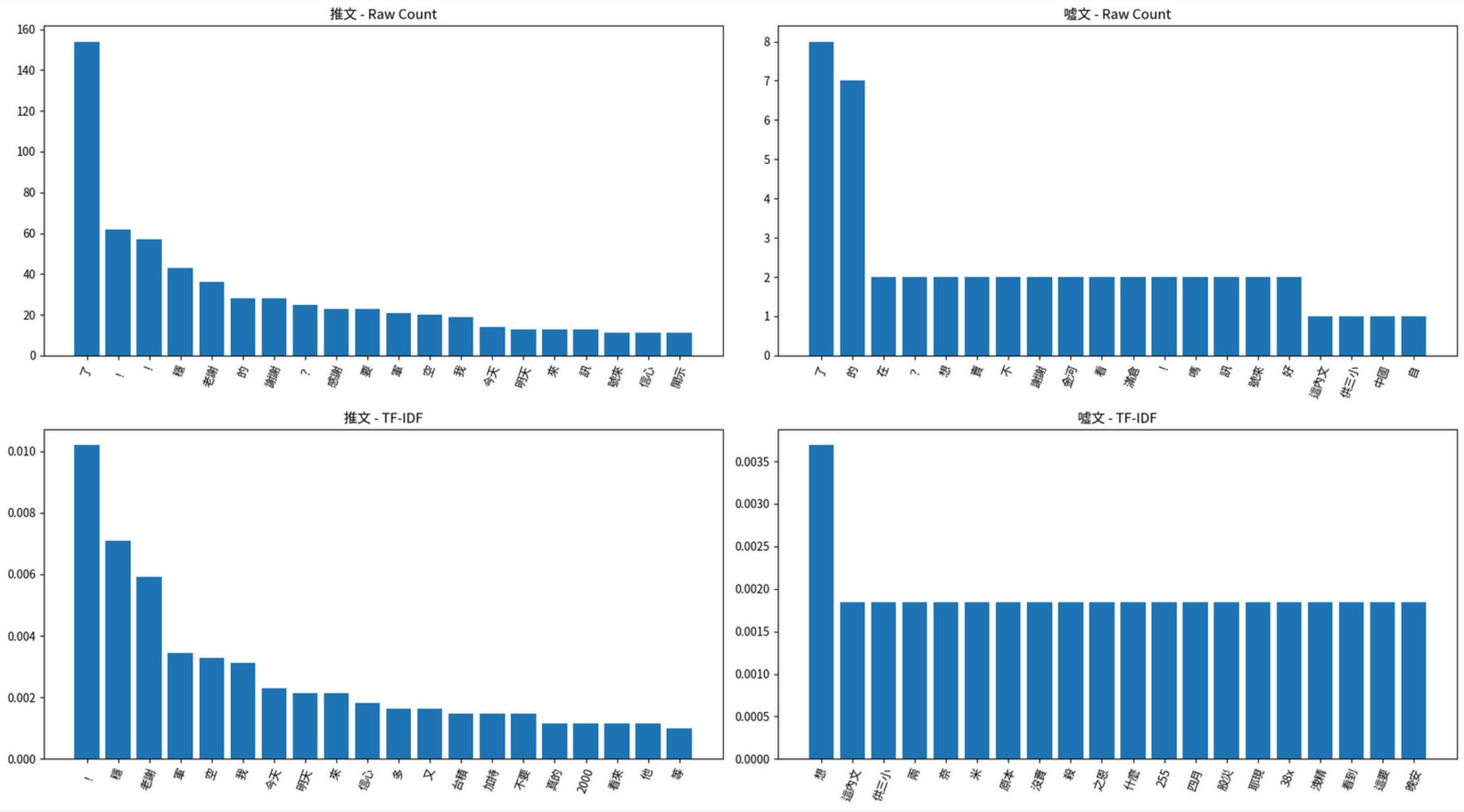
Applied TF-IDF to identify significant keywords in positive and negative comments.



RESULTS:

Before TF-IDF:
Results were dominated by symbols and generic words with no significant power.

After TF-IDF:

- **Positive posts:** terms like “stable”, “confidence”.
- **Negative posts:** terms like “what’s the point?”, “stock market crash”.

PERCEPTRON & LOGISTIC REGRESSION

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INPUT PROCESSING

- Chinese text is tokenized using jieba.
- Text is transformed into numerical features using TF-IDF vectors.
- Maximum of 5,000 features to capture relevant keywords while avoiding noise.

DATA SPLITING

- Dataset split into:

1. 80% Training set
2. 20% Testing set

- Random seed fixed to ensure reproducibility.

MODELS

- **Perceptron:**

Linear classifier based on a decision boundary.
Makes hard class decisions.

- **Logistic Regression:**

Linear classifier that maps scores to probabilities .
More stable and interpretable for binary classification.

EVALUATION METRICS

Accuracy: Proportion of correct predictions.

Precision: TP among predicted positive samples. Measures ability to avoid false positives.

Recall: Proportion of actual positives predicted correctly by the model.

VISUALIZATION

CONFUSION MATRIX:

- Helps identify model bias toward a class.
- Reveals class imbalance in the dataset.

RESULTS

- **Logistic Regression:**

Accuracy, Precision, Recall = 1.0
No misclassified samples.

- **Perceptron:**

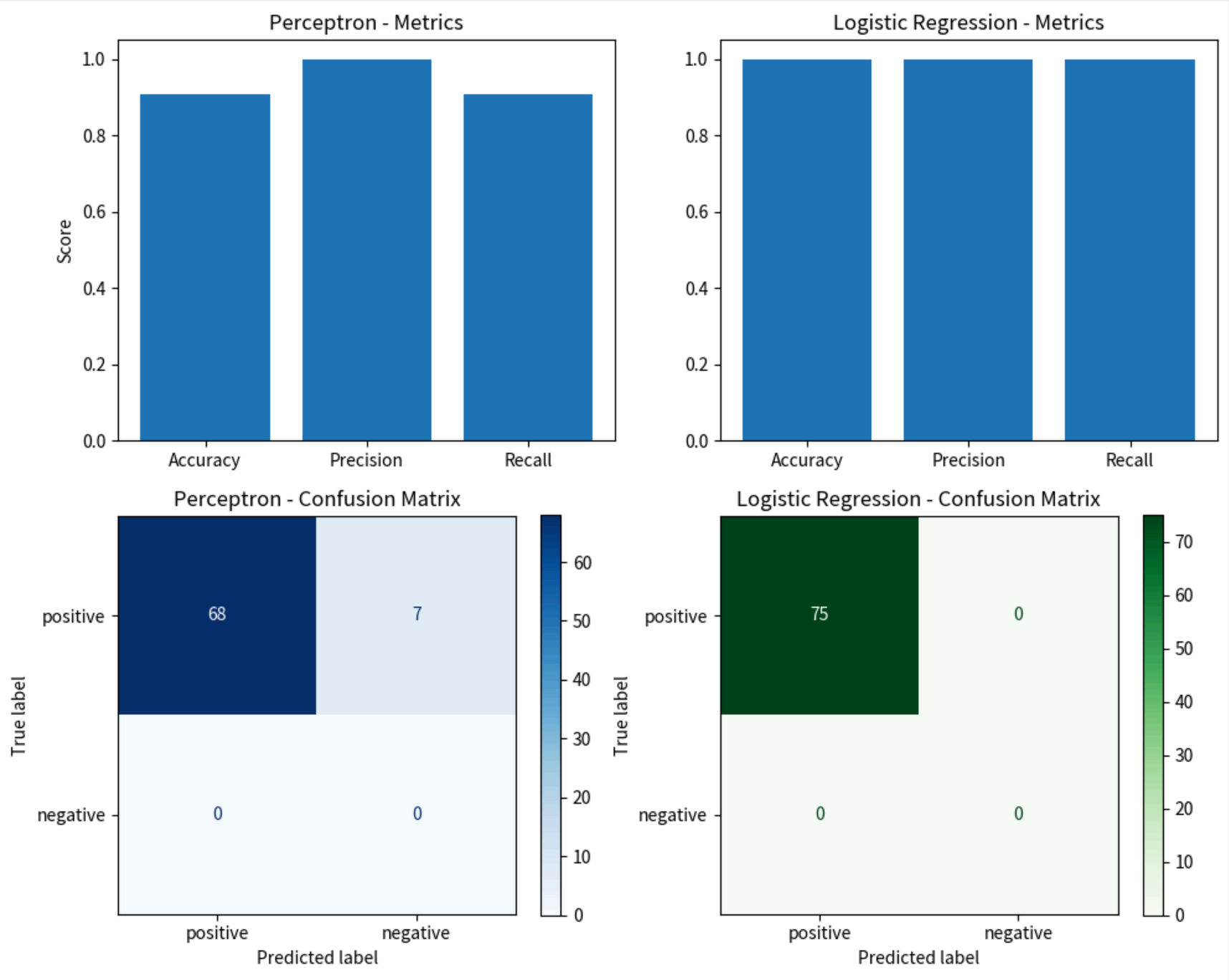
Slightly lower accuracy and recall. Misclassified a small number of positive samples.

LIMITATIONS

- No negative samples appeared in the test set.
- Class imbalance limits evaluation of negative class.
- Results may not generalize without more balanced data.

CONCLUSIONS

- TF-IDF features capture sentiment related words.
- Both models demonstrate feasibility of sentiment classification.
- Logistic Regression shows higher consistency, while Perceptron exhibits minor classification deviation.


Perceptron - Metrics
Score
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
Precision
Recall
Logistic Regression - Metrics
Perceptron - Confusion Matrix
True label
positive
negative
68
7
0
0
Predicted label
Logistic Regression - Confusion Matrix
75
0
0
0

SENTIMENT ANALYSIS WITH SNOW NLP

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- **SnowNLP is a sentiment analysis tool designed for Chinese text.**

Outputs a sentiment score between 0 and 1:

Close to 1: Positive sentiment
Close to 0: Negative sentiment
Around 0.5: Neutral, weak sentiment

PURPOSE

Check the level of consistency between the positive/ negative comments and the actual sentiment of the text.

Reclassification

Convert SnowNLP scores into labels:

- Score > 0.6 - Positive
- Score < 0.4 - Negative
- Otherwise - Neutral

Compare predicted sentiment with original PTT labels.

RESULTS

POSITIVE

- Many comments contain:

Complaints
Sarcasm
Irony

- Pressing 推 does not guarantee positive emotion.

NEGATIVE

- Sentiment is split among: Positive, neutral, negative.

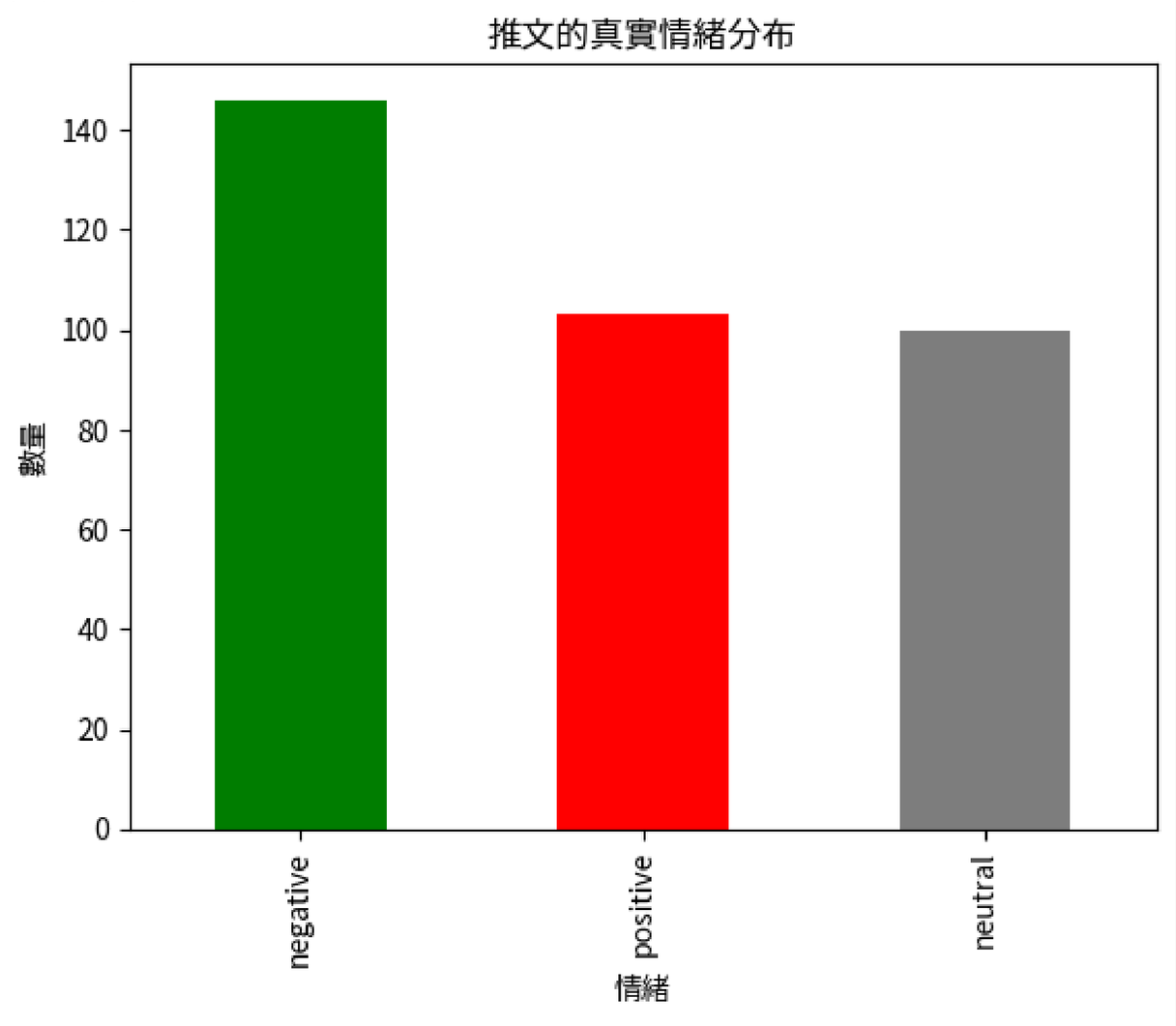
- Many 噓 comments are:

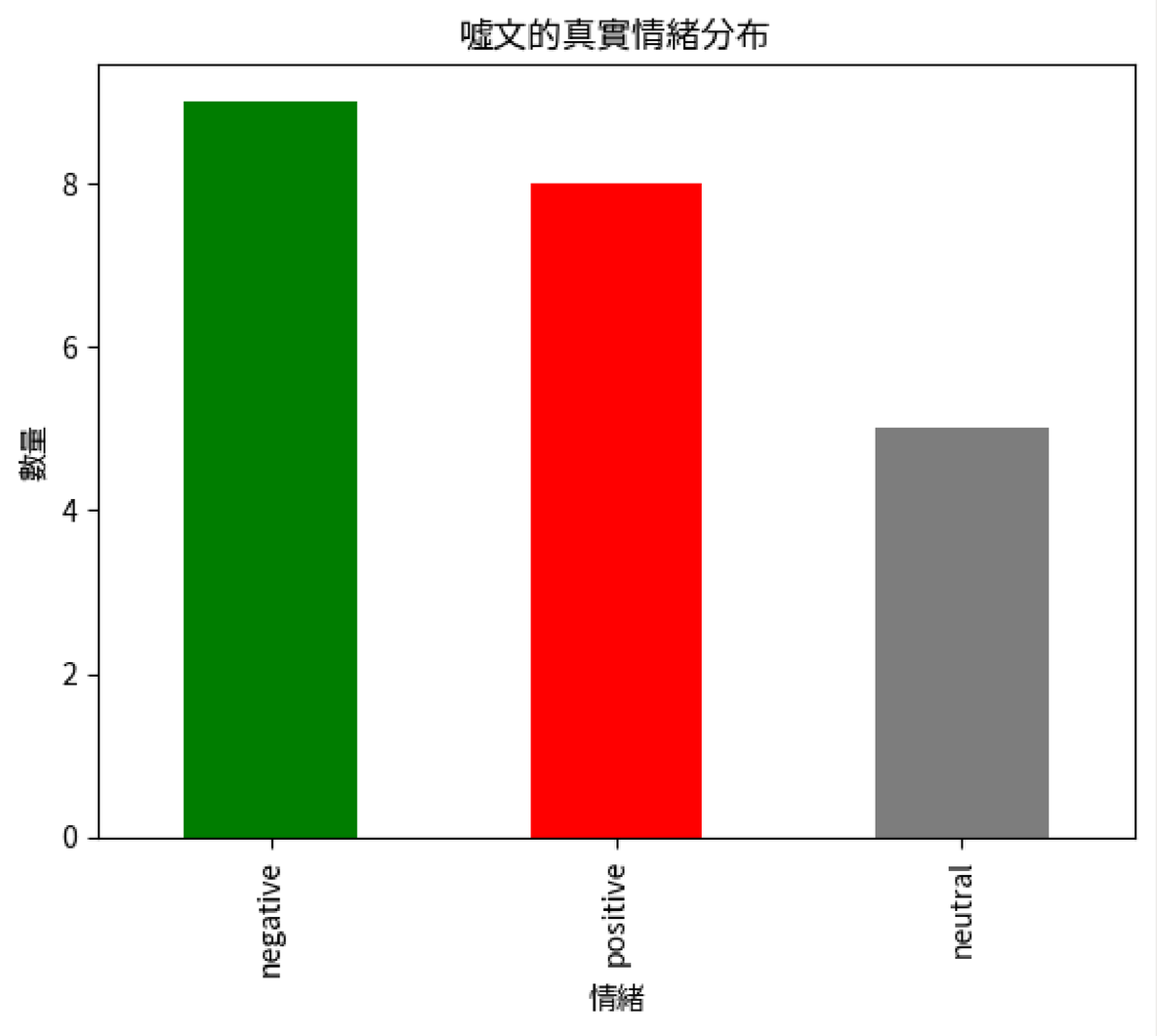
Neutral statements
Memes or teasing
Disagreement

CONCLUSION

PTT interaction labels do not reflect emotional sentiment. Many comments marked as “推” contain negative sentiment, some users may push while expressing criticism or sarcasm. Similarly, comments labeled “噓” show a mix of negative, neutral, and positive sentiments, suggesting that they use the labels to engage in discussion, joke, or tease.

Overall, “推” and “噓” function more as interaction or making a stance rather than accurate labels.


推文的真實情緒分布
數量
0
20
40
60
80
100
120
140
negative
positive
neutral
情緒


噓文的真實情緒分布
數量
0
2
4
6
8
negative
positive
neutral
情緒

Bibliography

- Choudhary, P. R. (2025). Sentiment analysis of social media data for public opinion and market trend prediction. International Journal of Research Publications in Engineering, Technology and Management (IJRPETM), 8(2), 1–10. https://doi.org/10.15662/IJRPETM.2025.0802002
- Global retail investors why don't make money? Study finds Taiwan retail investors lose up to 3.8% annually, three main reasons. (2024, August 14). China Times Economic Daily. https://www.ctee.com.tw/news/20240814700013-43100
- Heydarian, P., Bifet, A., & Corbet, S. (2024). Understanding market sentiment analysis: A survey. Journal of Economic Surveys, 39(5), 1125–1147. https://doi.org/10.1111/joes.12645
- Mokhtari, M., Seraj, A., Saeedi, N., & Karshenas, A. (2023). The impact of Twitter sentiments on stock market trends. Isfahan University of Technology, Department of Electrical and Computer Engineering. https://arxiv.org/pdf/2302.07244
- Picasso, A., Merello, S., Ma, Y., Oneto, L., & Cambria, E. (2019). Technical analysis and sentiment embeddings for market trend prediction. Expert Systems With Applications, 135, 60–70. https://doi.org/10.1016/j.eswa.2019.06.003
- Similarweb Ltd. (2025). Traffic sources and competitors of ptt.cc. Retrieved November 2025, from https://www.similarweb.com
- Tax statistics show securities transaction tax contributed 20 reds! Last year’s tax revenue reached NT$3.7 trillion, exceeding budget by NT$497.2 billion — second highest in history. (2025, January 13). SinoPac Securities RichClub. https://www.sinotrade.com.tw/richclub/news/6784dd1a31c1ba247ac9ff2d

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THANK YOU